Problem set -1

Macroeconomics 1

Hossein Joshaghani

Due: Optional

Send your answers to ravanmh@gmail.com.

Problem 1

In this problem, you will become acquainted with simple dynamical systems and create hand-drawn phase diagrams for them. For each of the following systems, do as follows:

- Find the fixed points.
- Determine their types.
- Plot separating manifolds of the system.
- Complete the phase diagram.

1. $\dot{x} = x(2 - x - y)\,,\ \dot{y} = x - y$
2. $\dot{x} = x - y\,,\ \dot{y} = 1 - e^x$
3. $\dot{x} = x^2 - y\,,\ \dot{y} = x - y$
4. $\dot{x} = y\,,\ \dot{y} = -x + y(1 - x^2)$

Problem 2

The 4th system from the first problem is known as "van der Pol oscillator". Use Python, Matlab, or any programming language of your choice to plot its phase diagram.